Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/02/2017
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
PI Address 1:	Fire Protection Engineering		
PI Address 2:	3104G Jm Patterson		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John (FM Global)		
Grant/Contract No.:	NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	One journal paper was published in 2017 [Ed. note: see previous year's Bibliography section] and a second one was submitted. Four conference papers or abstracts were published and presented. The detailed calorimeter model has improved. This considers a heat balance for the copper slug of the burner and may provide more accurate measurements of heat feedback to the surface than the heat flux gages. This model has been validated for the UMD (University of Maryland) burners by conducting normal gravity tests. The model was also applied to drop-tower burn tests. We are able to predict the average heat flux for these tests. This alternative approach to calculate heat flux is compared to the theoretical heat flux predicted by the analytical model of H. Baum. A heated sphere transient model was developed numerically and analytically (with Howard Baum) for the BRE burner. Two methods were tested to estimate the convective heat transfer coefficient (during drop tower tests) and compared to theoretical values. We are preparing an abstract for the National Combustion Meeting and a journal paper for Combustion and Flame. The solution for the 1D transient combustion model has been derived. The solution contains expressions for the velocity field, temperature field, density and species mixture fractions. The unsteady combustion model for the BRE burner is complete with the velocity field, temperature field, density and species mixture fractions derived. The model considers a porous ellipsoidal particle with fuel leaving its surface at a constant rate. The model has been tested against the 5 s drop tests and shows good agreement. James Quintiere gave a plenary lecture at the 9th International Seminar on Flame Structure, Novosibirsk, Russia. Its title was, A Gas Burner Emulator for Condensed-Phase Burning. Alex Snegirev’s team in St. Petersburg has completed their first CFD simulations of BRE flames. The UMD unsteady analytical model is being compared to the CFD simulations. The model is being modified to introduce radiative loss fraction as an additional parameter. Akshit Markan was awarded the UMD Graduate School’s Outstanding Graduate Assistant Award for 2017. This was for his role as Graduate Research Assistant on the BRE project. Only the top 2% of Graduate Assistants were selected for this award.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Abstracts for Journals and Proceedings	Markan A, Sunderland PB, Quintiere JG, de Ris JL, Stocker DP, Baum HR. "Unsteady Analysis and an Alternative Heat Flux Measurement Technique for the Burning Rate Emulator." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Awards	Markan A. (Akshit Markan) "UMD (University of Maryland) Graduate School’s Outstanding Graduate Assistant Award, May 2017." May-2017

Papers from Meeting Proceedings	Markan A, Quintiere JG, Sunderland P, de Ris JL, Stocker D. "Emulation of Condensed Fuel Flames using a Burning Rate Emulator (BRE) in Microgravity." 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 2017 10th US National Combustion Meeting, College Park, MD, April 23-26, 2017. 6 p. , Apr-2017
Papers from Meeting Proceedings	Quintiere JG, Markan A, Sunderland PB, Baum HR, deRis JL, Stocker DP, Lundstrom FV. "A Gas Burner Emulator for Condensed-Phase Burning." Ninth International Seminar on Flame Structure, Novosibirsk, Russia, July 10-14, 2017. Ninth International Seminar on Flame Structure, Novosibirsk, Russia, July 10-14, 2017. , Jul-2017